

# Supplemental Material for: Non-additive Strain–Dopant Energetics in Quasi-Hexagonal C<sub>60</sub> Graphullerene

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## I. FINITE-DIFFERENCE DERIVATION OF THE FOUR-CORNER COUPLING ESTIMATOR

This appendix derives the four-corner coupling estimator  $\mathcal{S}$  used in the main text [Eqs. (2)–(3)] from a Taylor expansion of total energy about the pristine reference, justifies the stencil choice against alternatives, and states symmetry, gauge, and error-propagation properties.

### A. Taylor expansion and discrete composition

Let  $E(\epsilon, \delta)$  denote the total energy of a supercell with biaxial strain parameter  $\epsilon$  and composition label  $\delta$  (pristine reference vs. one substitutional dopant per C<sub>60</sub> on the fixed production map). Although  $\delta$  is sampled on a discrete chemical grid (B, N, or P) rather than a continuous concentration field, the expansion below is the standard *finite-difference* representation of the underlying composition-dependent energy functional  $E(\epsilon, \delta)$  on the audited  $(\epsilon, \delta)$  stencil—not a claim that dopant concentration is a differentiable continuum variable.

Expand about  $(\epsilon_0, \delta_0) = (0, 0)$  to second order in increments  $(\Delta\epsilon, \Delta\delta)$ :

$$E(\epsilon_0 + \Delta\epsilon, \delta_0 + \Delta\delta) = E_0 + E_\epsilon \Delta\epsilon + E_\delta \Delta\delta + E_{\epsilon\delta} \Delta\epsilon \Delta\delta + \mathcal{O}(\Delta\epsilon^2, \Delta\delta^2, \Delta\epsilon^2 \Delta\delta, \Delta\epsilon \Delta\delta^2), \quad (1)$$

where  $E_\epsilon \equiv \partial E / \partial \epsilon|_{0,0}$ ,  $E_\delta \equiv \partial E / \partial \delta|_{0,0}$ , and  $E_{\epsilon\delta} \equiv \partial^2 E / \partial \epsilon \partial \delta|_{0,0}$ . The bilinear coefficient  $E_{\epsilon\delta}$  (denoted  $\gamma$  in continuum discussions) characterizes the intrinsic mixed energetic response; it is a *material functional* of the host, boundary conditions, and exchange–correlation approximation. Eq. (1) requires sufficiently small perturbations and a single, smooth energy branch; it fails across phase transitions or when strain drives large bond rearrangements (main text, Sec. IV E).

### B. Four-corner stencil and cancellation of linear terms

Define the four corner totals on the rectangular grid  $(\epsilon, \delta) \in \{(0, 0), (\Delta\epsilon, 0), (0, \Delta\delta), (\Delta\epsilon, \Delta\delta)\}$ :  $E_{00}$ ,  $E_{\epsilon 0}$ ,  $E_{0\delta}$ ,  $E_{\epsilon\delta}$ . The symmetric combination

$$\Delta E_{\text{cross}} \equiv E_{\epsilon\delta} - E_{\epsilon 0} - E_{0\delta} + E_{00} \quad (2)$$

substitutes Eq. (1) at each corner. All first-order terms cancel identically:

$$\Delta E_{\text{cross}} = E_{\epsilon\delta} \Delta\epsilon \Delta\delta + \mathcal{O}(\Delta\epsilon^2, \Delta\delta^2, \Delta\epsilon^2 \Delta\delta, \Delta\epsilon \Delta\delta^2). \quad (3)$$

Dividing by  $N_{\text{atoms}}$  gives the per-atom estimator  $\mathcal{S} = \Delta E_{\text{cross}} / N_{\text{atoms}}$  [Eq. (3)], i.e.,

$$\mathcal{S} \approx \frac{E_{\epsilon\delta}}{N_{\text{atoms}}} \Delta\epsilon \Delta\delta + \mathcal{O}(\Delta\epsilon^2, \Delta\delta^2, \Delta\epsilon^2 \Delta\delta, \Delta\epsilon \Delta\delta^2), \quad (4)$$

which is Eq. (4) of the main text. Thus  $\mathcal{S}$  *estimates* the mixed energetic response under finite  $(\Delta\epsilon, \Delta\delta)$ ; it is not itself the continuum coefficient  $\gamma \equiv E_{\epsilon\delta} / N_{\text{atoms}}$  unless the higher-order remainder is negligible.

The additive-screening bias is, by construction,  $\mathcal{S} = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)]$  [Eq. (2)], where each  $\Delta E$  is measured relative to the same pristine reference  $E(0, 0)$ .

### C. Why this four-point stencil?

The symmetric four-corner form is *not* unique, but it is the minimal second-order-accurate stencil on a rectangular  $(\epsilon, \delta)$  grid that (i) requires only four DFT totals, (ii) cancels all first-order strain and composition increments and any shared reference offset, and (iii) isolates the bilinear cross term in one algebraic step. Alternatives differ in cost, bias, or interpretability:

- **Central (five-point) stencils** on a finer grid reduce truncation error at the price of additional corners and, for discrete  $\delta$ , may interpolate between non-physical composition labels.
- **Least-squares fits** of  $E(\epsilon, \delta)$  to a low-order polynomial over many points can estimate  $E_{\epsilon\delta}$  but mix truncation error with fit-window choice and do not map one-to-one onto the additive-screening bias audited here.
- **Full mixed Hessian**  $\partial^2 E / \partial \epsilon \partial \delta$  from analytic or numerical derivatives of relaxed energies requires continuous, well-defined  $\delta$  and is ill posed on the present discrete substitution grid.

Asymmetric three-point schemes retain linear contamination in at least one channel; Table II contrasts  $\mathcal{S}$  with other descriptors.

### D. Symmetry, gauge invariance, and units

*a. Symmetry (Clairaut).* If  $E_{\epsilon\delta} = E_{\delta\epsilon}$  (Schwarz/Clairaut theorem for a smooth  $E$ ), the symmetric stencil (2) is invariant under exchanging the

TABLE I. Positioning relative to representative prior work on qHP graphullerene and related covalent networks (not an exhaustive literature map). “Joint  $(\epsilon, \delta)$ ” denotes simultaneous strain and substitutional doping in the same calculation series; “four-corner total-energy audit” denotes symmetric  $(\epsilon, \delta)$  corner differences as in the main-text definition of  $\mathcal{S}$ .

| Prior focus                                                | Joint $(\epsilon, \delta)$ |
|------------------------------------------------------------|----------------------------|
| Separate strain or doping maps [1–5]                       | ×                          |
| Nonlinear band-edge / transport response [6–8]             | Partial                    |
| Alloy cluster expansion (effective $c\epsilon x$ ) [9, 10] | ✓                          |
| This work (qHP C <sub>60</sub> , PBE+D3)                   | ✓                          |

order in which the  $\epsilon$  and  $\delta$  increments are applied on the grid: the estimated cross term is the same whether one views strain or composition as the “first” axis. This is a property of the *stencil*, not a claim that biaxial strain and chemical substitution are physically interchangeable controls (main text, Sec. II C).

*b. Gauge invariance.* Adding an arbitrary constant  $C$  to all four corner totals leaves  $\mathcal{S}$  unchanged; only energy *differences* among the four corners enter. Shared systematic shifts (e.g., uniform XC offsets) therefore cancel in the four-corner combination when all corners share the same computational protocol.

*c. Units.*  $\mathcal{S}$  is reported in meV per atom. The dimensionless additive-failure fraction  $\eta = |\mathcal{S}|/|\Delta E_{\text{coupled}}|$  [Eq. (6)] has no units. The continuum coefficient  $\gamma \equiv \partial^2 E / \partial \epsilon \partial \delta$  carries units of energy per atom per strain increment per composition increment; on the production grid ( $\Delta\epsilon = 3\%$ , discrete  $\Delta\delta$  from pristine to one substituent per C<sub>60</sub>),  $\mathcal{S} \approx \gamma \Delta\epsilon \Delta\delta$  up to higher-order remainders.

### E. Error propagation

With  $\mathcal{S} = (E_{\epsilon\delta} - E_{\epsilon 0} - E_{0\delta} + E_{00})/N_{\text{atoms}}$  and independent, equal per-atom uncertainties  $\sigma_e$  on each converged corner total,

$$\sigma_{\mathcal{S}}^2 = \frac{1}{N_{\text{atoms}}^2} (\sigma_e^2 + \sigma_e^2 + \sigma_e^2 + \sigma_e^2) = 4\sigma_e^2, \quad \sigma_{\mathcal{S}} = 2\sigma_e, \quad (5)$$

which is Eq. (5) of the main text when corners are uncorrelated. Correlated SCF noise across chemically related corners would reduce the effective coefficient below 2; the main-text reporting band uses  $2\sigma_e$  as a conservative upper bound (Table I).

## II. LITERATURE POSITIONING AND DESCRIPTOR COMPARISON

### III. ARCHIVED LEGACY-PBE TETRAMER BASELINE

Table III reproduces the archived tetramer cluster map (PBE without D3; Cartesian coordinates not affinely scaled) retained for reproducibility; it is *not* used for quantitative strain-physics claims in the main text.

### IV. CALCULATION INVENTORY (REPRODUCIBILITY)

Tables IV and V list the DFT groups underlying the main-text claims and superseded exploratory grids retained only for reproducibility. They are *not* scientific results tables.

### V. FULL TETRAMER $\pi$ -DOS AT $\epsilon=0$

### VI. PERIODIC FRONTIER-ORBITAL LOCALIZATION

Cage-resolved VBM/CBM isosurfaces on periodic  $n \times \text{C}_{60}$  cells ( $n = 2-8$ ) at  $\epsilon=0$  complement the tetramer  $\pi$ -DOS above and the main-text electronic taxonomy (donor/acceptor vs. size dilution). Carbon cages are shown as uniform gray MSMS molecular surfaces (probe 1.3 Å); heteroatoms use element colors.

### VII. PERIODIC COUPLING-ENERGY ROWS ( $n \geq 6$ )

### VIII. FOUR-CORNER TOTAL ENERGIES (+3%)

### IX. MARCUS REORGANIZATION ENERGIES

Polaron transport diagnostics are reported here for completeness; the main-text analysis centers on  $\mathcal{S}$ .

### X. HOPPING INTEGRAL $J$

On the tetramer dimer grid, B substitution at  $\epsilon=0$  delocalizes the charged-state orbital (IPR 75.0  $\rightarrow$  24.6) while raising  $J$  from 26.6 to 46.4 meV; coupled B @ +3% gives  $J = 37.2$  meV and IPR 23.1. The criterion  $J > \lambda/2$  for polaron hopping dominance [4] is *not* met at PBE+D3 ( $J < \lambda/2$ ), so a polaron transition is not confirmed on this grid.

TABLE II. Positioning of the strain-dopant coupling energy  $\mathcal{S}$  (main-text four-corner estimator) against common stability and electronic descriptors. Algebraically  $\mathcal{S}$  has the form of an interaction (coupling) energy; the distinction is *dimensional*, i.e. which axes of the joint  $(\epsilon, \delta)$  control space each quantity resolves. “Additive-screening error” denotes the bias incurred when total energies under joint load are assembled additively from separate strain-only and doping-only scans;  $\mathcal{S}$  equals that bias by construction. “Double-reference cross term” denotes a symmetric difference of differences over the four  $(\epsilon, \delta)$  corners rather than subtraction of a single reference state or fragment pair. Entries are conceptual scope, not numerical performance; consistent with its energetic definition,  $|\mathcal{S}|$  is *not* reducible to a single geometric ( $\Delta\bar{d}$ ) or charge ( $\Delta q$ ) channel alone.

| Descriptor                                           | Nonlinear<br>( $\epsilon, \delta$ ) coupling | Additive-screening<br>error | Double-reference<br>cross term | Joint strain<br>+ doping design |
|------------------------------------------------------|----------------------------------------------|-----------------------------|--------------------------------|---------------------------------|
| Formation / substitution energy $E_{\text{sub}}$     | ×                                            | ×                           | ×                              | ×                               |
| Binding / interaction energy $\Delta E_{\text{int}}$ | Partial                                      | ×                           | ×                              | Partial                         |
| Linear strain coefficient $\alpha$                   | ×                                            | ×                           | ×                              | Partial                         |
| Band gap $E_g$ / work function                       | ×                                            | ×                           | ×                              | ×                               |
| $\mathcal{S}$ (this work)                            | ✓                                            | ✓                           | ✓                              | ✓                               |

TABLE III. Archived legacy-PBE (no D3) tetramer cluster map retained for reproducibility only; not used for main-text quantitative claims. Cartesian coordinates are not affinely scaled.  $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$  at  $\epsilon=0$  (no chemical potentials; relative ranking within this protocol only).

| System   | $n_{\text{dop}}$ | $E(\epsilon=0)$ ( $E_h$ ) | $E(\epsilon=+5\%)$ ( $E_h$ ) | $\alpha$ (meV/%) | $E_{\text{sub}}$ (eV/dopant) |
|----------|------------------|---------------------------|------------------------------|------------------|------------------------------|
| Pristine | —                | −1363.5512                | −1363.5509                   | −1.0             | —                            |
| B-doped  | 13               | −1328.4900                | −1328.4090                   | +56.9            | +73.4                        |
| N-doped  | 12               | −1414.3812                | −1414.4053                   | −303.2           | −115.3                       |
| P-doped  | 12               | −1370.4965                | −1370.4888                   | −19.1            | −15.8                        |

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TABLE IV. DFT calculation groups supporting the main-text claims (reproducibility inventory; Supplemental Material only). Superseded exploratory grids appear in Table V.

| Calculation group              | System                         | Count | Protocol                                    |
|--------------------------------|--------------------------------|-------|---------------------------------------------|
| Archived baseline (legacy PBE) | $4 \times \text{C}_{60}$       | 24    | Rigid strain; not used for main-text claims |
| $\pi$ -DOS and gaps            | $4 \times \text{C}_{60}$       | 12    | PBE+D3                                      |
| IPR (polaron dimer)            | Dimer                          | 4     | Pristine/B; $\epsilon=0, +3\%$              |
| Hopping $J$                    | Dimer                          | 4     | PBE+D3                                      |
| Charged polarons <sup>a</sup>  | $4 \times \text{C}_{60}$       | 20    | UKS; geometry opt. + vertical SP            |
| Periodic size scaling          | $n \times \text{C}_{60}$       | 41    | PBE+D3; 400/350 Ry                          |
| Cutoff benchmark               | $6 \times \text{C}_{60}$ N     | 1     | +3%; 400 Ry single-point                    |
| Ionic relaxation               | $4 \times \text{C}_{60}$       | 4     | Fixed-cell GEO; $P$ corners                 |
| Ionic relaxation               | $1 \times \text{C}_{60}$ P     | 4     | Fixed-cell GEO; periodic                    |
| Alternate placement            | $4 \times \text{C}_{60}$       | 24    | Rigid PBE+D3                                |
| Periodic placement             | $4 \times \text{C}_{60}$       | 16    | $n=4$ ; two independent maps                |
| Matched-functional rigid       | $4 \times \text{C}_{60}$       | 4     | PBE+D3 at relaxed coordinates               |
| Reference placement            | $4 \times \text{C}_{60}$       | 24    | Rigid PBE+D3                                |
| Hirshfeld population           | $1 \times \text{C}_{60}$ B/N/P | 18    | Spin-polarized KS                           |

<sup>a</sup>Six relaxed tetramer geometries provide charged-state initial structures.

TABLE V. Exploratory DFT grids superseded by the production protocol (reproducibility archive only).

| Task group              | System                   | $N_{\text{conv}}$ | Disposition  |
|-------------------------|--------------------------|-------------------|--------------|
| Early strain probe      | $2 \times \text{C}_{60}$ | 5                 | Superseded   |
| Single-molecule doping  | $\text{C}_{60}$          | 11                | Superseded   |
| Early electronics       | $2 \times \text{C}_{60}$ | 24                | Superseded   |
| Exploratory design grid | $4 \times \text{C}_{60}$ | 20                | Archive only |

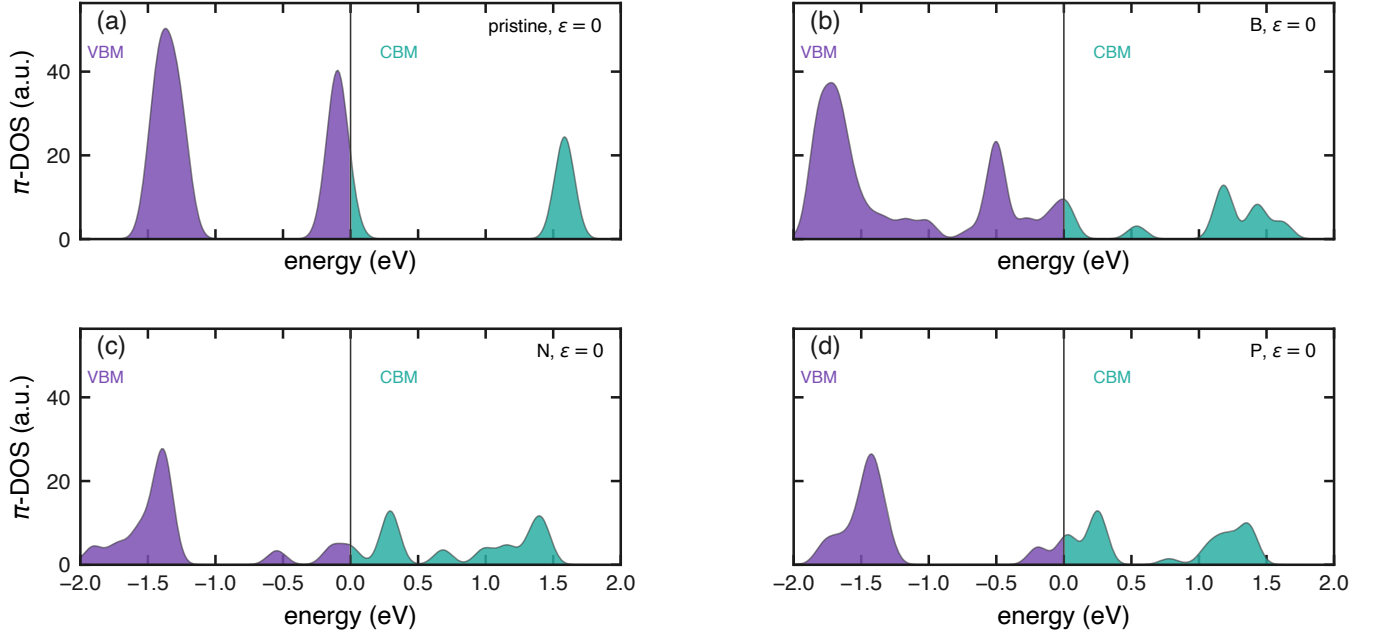
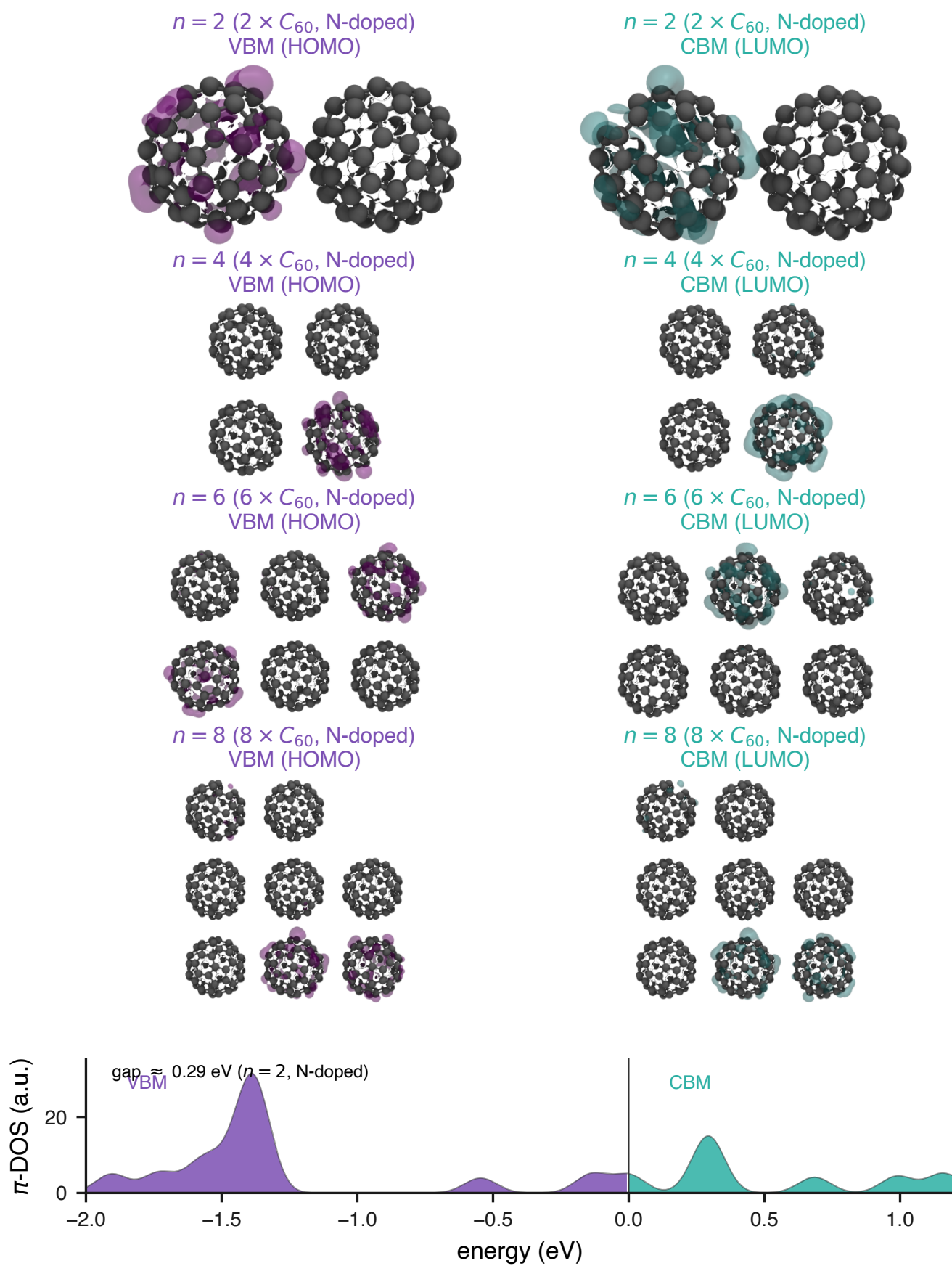


FIG. 1. Full  $\pi$ -DOS panels for pristine and B/N/P-doped tetramers at  $\epsilon=0$  (PBE+D3), complementary to the main-text electronic-structure figure.

qHP  $C_{60}$  graphullerene ( $\varepsilon = 0\%$ , N-doped)

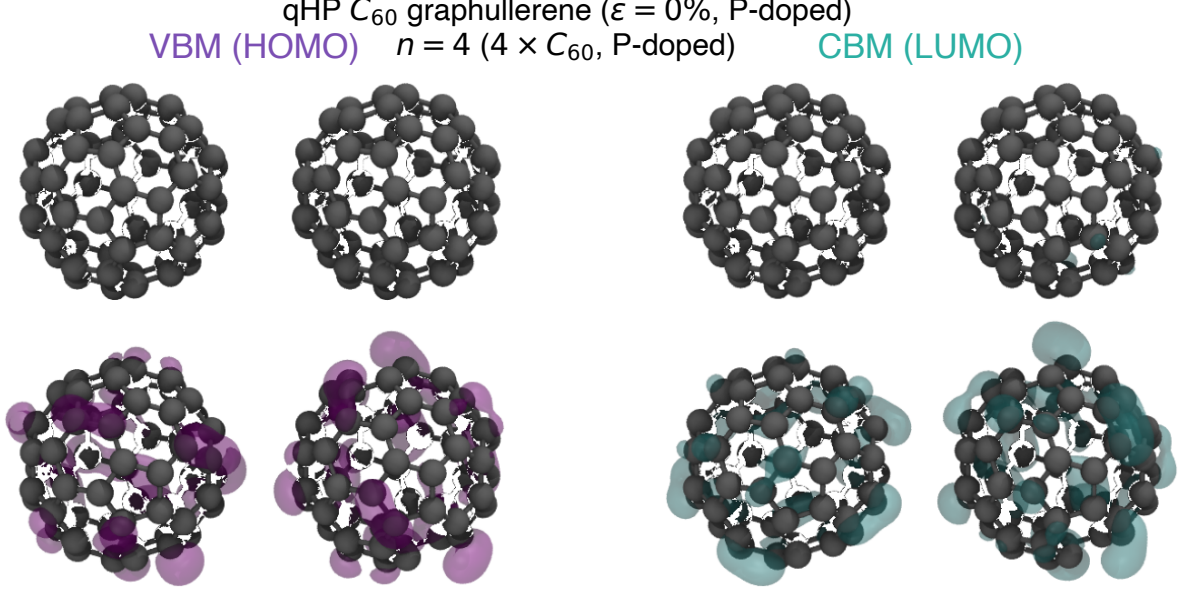


FIG. 3. P-doped  $n=4$  supercell at  $\epsilon=0$ : VBM (left) and CBM (right) on gray MSMS cages. Both frontier states localize on a subset of cages, providing a spatial counterpart to the large periodic  $|S|$  of size-mismatched P reported in the main text.

TABLE VI. Supplemental periodic coupling-energy rows at +3% ( $n \geq 6$ ; 350 Ry production cutoff; completeness only).

| $n$ | Dopant | $\epsilon$ (%) | $\mathcal{S}$ (meV/atom) |
|-----|--------|----------------|--------------------------|
| 6   | B      | +3             | -7.76                    |
| 6   | N      | +3             | +2.38                    |
| 6   | P      | +3             | -31.89                   |
| 8   | B      | +3             | -11.26                   |
| 8   | N      | +3             | -2.30                    |
| 8   | P      | +3             | -29.94                   |

TABLE VII. Machine-readable four-corner total energies per atom (Ha/atom) underlying  $\mathcal{S}$  at +3% biaxial strain (PBE+D3, rigid protocol, reference placement seed 42). Corners follow Eq. (3):  $E_{00} = E(0,0)$ ,  $E_{e0} = E(\Delta\epsilon,0)$ ,  $E_{0\delta} = E(0,\Delta\delta)$ ,  $E_{e\delta} = E(\Delta\epsilon,\Delta\delta)$ , all per atom relative to the same supercell size  $n \times C_{60}$ .  $\mathcal{S}$  (meV/atom) matches Eq. (2) and the open audit chain (CSV mirror in the public data release).

| $n$ | $\delta$ | $\epsilon$ (%) | $E_{00}$      | $E_{e0}$      | $E_{0\delta}$ | $E_{e\delta}$ | $\mathcal{S}$ |
|-----|----------|----------------|---------------|---------------|---------------|---------------|---------------|
| 1   | B        | +3             | -5.6826713370 | -5.6825614149 | -5.6339765778 | -5.6341744697 | -8.38         |
| 1   | N        | +3             | -5.6826713370 | -5.6825614149 | -5.7532834847 | -5.7530890722 | +2.30         |
| 1   | P        | +3             | -5.6826713370 | -5.6825614149 | -5.6922302079 | -5.6932928189 | -31.91        |
| 2   | B        | +3             | -5.6817028549 | -5.6817462745 | -5.6338466634 | -5.6339702803 | -2.18         |
| 2   | N        | +3             | -5.6817028549 | -5.6817462745 | -5.7532007583 | -5.7529924491 | +6.85         |
| 2   | P        | +3             | -5.6817028549 | -5.6817462745 | -5.6895002016 | -5.6905831259 | -28.29        |
| 4   | B        | +3             | -5.6828203886 | -5.6827081746 | -5.6341906516 | -5.6343073761 | -6.23         |
| 4   | N        | +3             | -5.6828203886 | -5.6827081746 | -5.7535095848 | -5.7532889091 | +2.95         |
| 4   | P        | +3             | -5.6828203886 | -5.6827081746 | -5.6924497414 | -5.6932089910 | -23.71        |
| 6   | B        | +3             | -5.6828517956 | -5.6827379495 | -5.6340195050 | -5.6341908867 | -7.76         |
| 6   | N        | +3             | -5.6828517956 | -5.6827379495 | -5.7533481235 | -5.7531468231 | +2.38         |
| 6   | P        | +3             | -5.6828517956 | -5.6827379495 | -5.6916511984 | -5.6927091482 | -31.89        |
| 8   | B        | +3             | -5.6828645675 | -5.6825524079 | -5.6341364256 | -5.6342380770 | -11.26        |
| 8   | N        | +3             | -5.6828645675 | -5.6825524079 | -5.7534470201 | -5.7532194756 | -2.30         |
| 8   | P        | +3             | -5.6828645675 | -5.6825524079 | -5.6919097127 | -5.6926978180 | -29.94        |

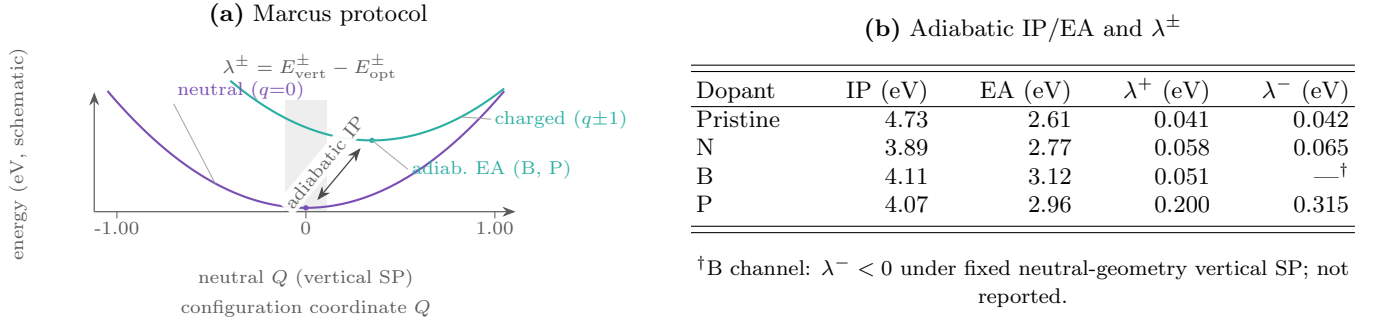


FIG. 4. Marcus reorganization energies from charged-state geometry optimizations and vertical single-points at the neutral geometry. Panel (a) schematically defines the vertical-adiabatic decomposition following Ref. 4; panel (b) tabulates adiabatic IP/EA and  $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$  for pristine and B/N/P tetramers. The electron-channel  $\lambda^-$  for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

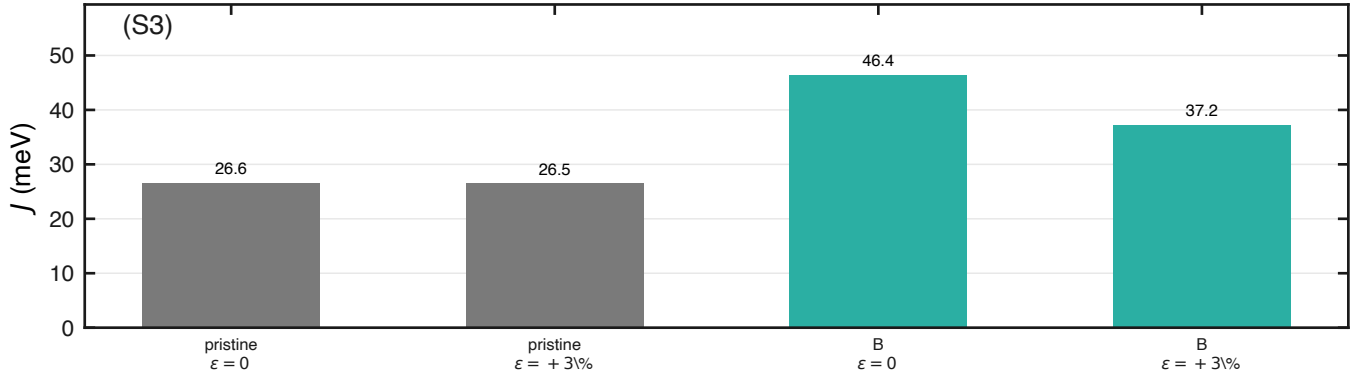


FIG. 5. Nearest-neighbor electronic coupling  $J$  on the tetramer dimer grid (pristine and B-doped cells at  $\epsilon=0$  and  $+3\%$ ; PBE+D3). Companion inverse-participation-ratio values for the same four corners are tabulated in the open data release (Methods).